

BLOCKCHAIN OLYMPIAD BANGLADESH 2026

ARTIFICIAL INTELLIGENCE CATEGORY

SHEBAR JANALA

Voice-first, explainable AI for government-service access and local accountability in Bangladesh*AccessAI is the intelligent service-discovery feature within the Shebar Janala platform.***Central thesis**

Citizens should not need to know a programme name before government can help them. Shebar Janala lets a person describe a life event in ordinary Bangla, identifies relevant support, checks eligibility through auditable rules, explains the result with AI, guides the next steps, and creates an accountability path when delivery fails.

2

district pilot

33

citizens engaged

102

complaints captured

685automated tests in current
AccessAI codebase**Author****Tanjim Mahmud Rakin**Corresponding author: tanjimahmudwork@gmail.com[Live prototype](#)**Executive Abstract**

Bangladesh has expanded digital government channels, yet the citizen journey remains fragmented at the moment it matters most: a person facing job loss, disability, widowhood, illness, educational need, climate shock or financial hardship may not know which programme exists, whether they qualify, which documents are required, where to go, or what to do when service delivery fails. Shebar Janala addresses this end-to-end gap. Its AccessAI feature converts natural Bangla/English text or speech into a structured, explainable service-discovery journey. AI is used where statistical learning is genuinely necessary - speech recognition, semantic retrieval, information extraction and grounded language rendering - while consequential eligibility decisions remain deterministic, replayable and auditable. The broader platform then links discovery to action plans, deadlines, nearby services via OpenStreetMap, and citizen complaint/accountability workflows. A working prototype already implements the core eligibility, retrieval, tracking, mapping, administration and safety architecture. Our next technical step is a domain-adapted Bangla civic AI stack, trained/evaluated on consented and public data with explicit abstention, subgroup evaluation and human confirmation. The design goal is not a fluent chatbot; it is a low-cost, voice-accessible public-service navigation layer whose answers are traceable and whose failures are visible.

1. Vision & Problem Statement

Vision. A citizen should be able to say what happened in their life and receive a trustworthy path to the right public support - without first knowing the agency, programme name, legal wording or office location.

The four-question gap

1) What support can I get? 2) Am I eligible and why? 3) What do I need and where do I go? 4) What happens if the service is not delivered?

1.1 Why this matters now

Bangladesh's National Social Security Strategy 2026-2035 explicitly continues the shift away from fragmented safety-net programmes toward a more coherent, lifecycle-oriented and institutionally coordinated system, while acknowledging persistent exclusion and targeting challenges [2]. MyGov already provides a one-stop channel for many government services and connects citizens to the national 333 help channel [3]. The remaining opportunity is not to replace those assets, but to add a citizen-intent layer that can understand a life event, evaluate structured eligibility, and connect the user to the correct service and accountability pathway.

1.2 The problem is a chain, not a missing website

Failure	What goes wrong
Discovery	Information is distributed across portals, circulars, offices and agencies. Citizens often need to know what to search for before they can search.
Interpretation	Eligibility can combine age, income, household, geography, occupation, disability and other conditions that are difficult to evaluate from prose.
Accessibility	Formal Bangla, long forms and typing-heavy interfaces exclude citizens who are more comfortable speaking than reading or typing.
Execution	Even after finding a programme, citizens may miss documents, deadlines, sequence or the correct physical service point.
Accountability	When service fails or local problems remain unresolved, discovery and complaint channels are disconnected; context is lost and follow-up is weak.

1.3 Primary stakeholders

Primary users are citizens in rural and underserved communities, particularly people with limited digital literacy, shared/low-specification devices or accessibility needs. Secondary users include Union Digital Centre operators, local government officials, programme administrators, NGO caseworkers and oversight teams. Each benefits from a structured, cited and replayable interpretation of citizen needs rather than ad-hoc advice.

Design principle

Do not optimize for the citizen who already knows the service name. Optimize for the citizen who only knows the problem they are living through.

2. Use Cases & Existing Solutions

2.1 Three connected use cases

Use case	Outcome
A. AccessAI - service discovery	Citizen describes a life event by text/voice. The system identifies relevant programmes, checks eligibility, explains why, lists documents and steps, creates an action plan, tracks deadlines and shows nearby service points.
B. Shebar Janala - local accountability	Citizen reports a local service failure or community issue using text, image, location and eventually voice. The platform supports moderation, routing, voting/flagging, status tracking and escalation.
C. Government / operator intelligence	Authorized staff maintain programme data and rules, review flagged content, monitor demand patterns, service gaps, complaint categories, data freshness and AI quality through role-based dashboards.

2.2 Why existing approaches are not enough

Alternative	Strength	Remaining gap
MyGov / portals	Authoritative service access	Best when the citizen already knows the service; limited life-event discovery and no cross-programme explainable eligibility layer.
333 / human helpline	Voice-first, trusted, low literacy barrier	Human capacity scales linearly; decisions are difficult to replay and compare across many numeric programme rules.
UDC / local operator	Physical trust and assisted service	Knowledge quality can vary; interpretation burden remains with the operator.
General AI assistant	Fluent natural-language interaction	A plausible threshold is dangerous when the citizen cannot distinguish plausible from authoritative; provenance and deterministic eligibility are missing.
Complaint-only portals	Creates a reporting channel	Does not solve upstream discovery/eligibility or retain a continuous citizen journey.

2.3 Evidence from the first Shebar Janala pilot

2 districts	33 citizens	102 complaints
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The pilot validated demand for a structured citizen-reporting channel, but it also exposed an important sustainability problem: a field-heavy, complaint-centric operating model was not cost-feasible at the scale we wanted. Instead of hiding that result, we used it to pivot. The current Shebar Janala combines the accountability layer with AccessAI so that the platform creates value before a complaint occurs - helping citizens discover support, avoid unnecessary trips and complete service journeys - while retaining reporting and escalation when delivery fails.

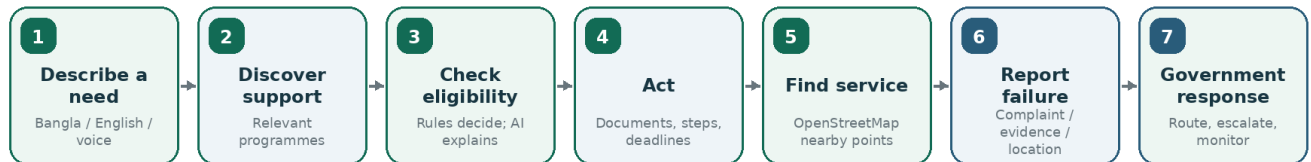
This pivot is a product-learning milestone, not a change of mission: the mission remains accessible and accountable public service; the mechanism is now more scalable and commercially viable.

3. Proposed AI Solution: Shebar Janala

Shebar Janala is the overall digital governance platform. AccessAI is its core AI-powered service-discovery feature. Together they create a continuous journey from citizen intent to government action and, when necessary, accountability.

Shebar Janala: one continuous citizen journey

AccessAI is the intelligent service-discovery feature inside the broader governance platform.



Closed feedback loop

The platform does not stop at “here is a link”. It follows the citizen from discovery to action, and if delivery fails, the same identity and context can create a structured accountability signal for local administration.

Figure 1. End-to-end citizen journey. The AI feature is embedded inside the broader Shebar Janala governance workflow.

3.1 Intended functionality

Function	What happens
Understand	Accept Bangla/English/Banglish text and voice; detect life event; extract candidate profile facts with provenance.
Discover	Search verified programme information using lexical and semantic retrieval plus location/life-event filters.
Decide safely	Use deterministic, versioned eligibility rules. Missing information becomes unknown, not an automatic rejection.
Explain	Generate a grounded, plain-language explanation from fixed facts; show citations, confidence and the decisive reason.
Execute	Produce document checklist, application steps, action plan, reminders and nearby service locations.
Escalate	If service fails, create a complaint/report in Shebar Janala and route it into local-government review and escalation.

3.2 What makes the solution different

- AI does not decide statutory eligibility. The system deliberately separates probabilistic understanding from deterministic public-service decisions.
- The product is designed around life events rather than ministry structure or programme names.
- It connects digital guidance to physical service points through OpenStreetMap/Overpass and then to complaint/accountability workflows.
- It degrades safely: if a model or provider is unavailable, deterministic search/composition remains usable and the system states the limitation.

4. AI Architecture & Methodology

The technical philosophy is simple: use machine learning only for open-ended perception or language tasks, and place a checkable gate between every model output and any consequential citizen fact.

AI architecture: models propose; deterministic and human gates decide

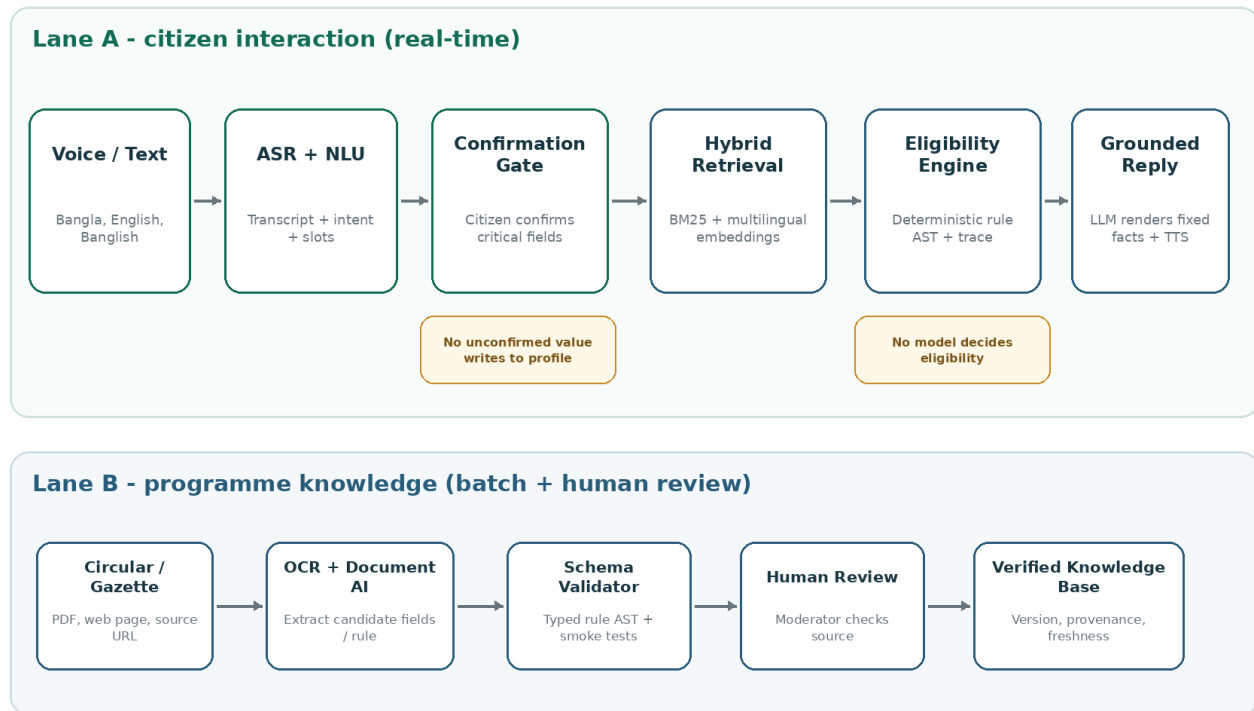


Figure 2. Two-lane AI architecture. No model has a direct path to an eligibility verdict or an unreviewed programme rule.

4.1 AI components

Capability	Role and safety boundary
A1. Bangla ASR / TTS	Speech recognition and read-aloud for low-literacy and hands-free use. Domain adaptation targets regional speech, noise and code-mixing.
A2. Slot extraction	Fine-tuned encoder extracts age, income, household, occupation, education, land, geography and other profile fields with calibrated abstention.
A3. Semantic retrieval	Multilingual embeddings complement BM25 so paraphrases can retrieve relevant programme text even with low lexical overlap.
A4. Document intelligence	OCR/layout + constrained extraction converts circulars into proposed structured rules. The output enters review; it never publishes automatically.
A5. Grounded generation	An LLM renders a pre-computed response plan into clear Bangla/English. It cannot introduce programmes, thresholds or benefits not in the plan.

Rules decide; AI understands and explains

This boundary is central to fairness, reproducibility and government trust. A model may propose a transcript, slot or draft rule; a citizen, validator or authorized reviewer confirms it before it becomes authoritative.

5. Data, Models, Training & Evaluation

5.1 Data sources

Source	Use and governance
Government / development programme data	Public websites, circulars, gazettes and official programme descriptions. Store source URL, retrieval date, verification status, version and reviewer.
Citizen profile data	Voluntarily provided eligibility fields. Sensitive fields are purpose-gated; missing information is allowed.
Voice / language data	Public Bangla corpora such as Common Voice and OpenSLR for adaptation, plus opt-in in-domain utterances and corrections; audio is discarded after transcription unless separate research consent is obtained.
Geospatial data	OpenStreetMap / Overpass for nearby service discovery, supplemented by curated service locations and provenance labels.
Pilot and feedback data	Complaints, helpfulness feedback, routing outcomes and corrections used primarily for evaluation and product improvement under governance controls.

5.2 Our model strategy

We are building a domain-adapted Bangla civic AI stack rather than training a general foundation model from scratch. The project controls the fine-tuned weights/adapters, task schemas, evaluation sets, prompts, abstention thresholds, release gates and model registry required for this specific public-service domain.

Layer	Model / method
Speech	Whisper-family baseline, domain-adapted on Bangla civic/service language; evaluate word/character error and critical-slot accuracy by district and demographic slice [4].
Extraction	BanglaBERT / XLM-R class encoder with span-and-slot heads; calibrated confidence and per-field abstention [5].
Retrieval	LaBSE or multilingual-E5 class embeddings fused with BM25 by Reciprocal Rank Fusion [6-8].
Generation	Hosted or self-hosted instruction model used only to render grounded facts; deterministic composer is fallback.
Document AI	Bangla OCR/layout model + schema-constrained LLM extraction into a closed rule grammar, followed by validation and human approval.

5.3 Evaluation gates before release

- Critical numeric fields (income, age, land/asset thresholds): target precision ≥ 0.98 ; otherwise abstain and ask the citizen to confirm.
- ASR: report WER/CER plus critical-slot accuracy by district, age and gender; a low average WER is not enough if numbers are corrupted.
- Retrieval: Recall@10, Precision@3 and nDCG@10 compared with BM25-only; semantic retrieval ships only if it improves recall without reducing top-result precision.
- Grounded generation: citation coverage, grounding-failure rate, human helpfulness/error review and deterministic fact comparison.

6. Architecture & Infrastructure

Platform architecture and deployment boundary

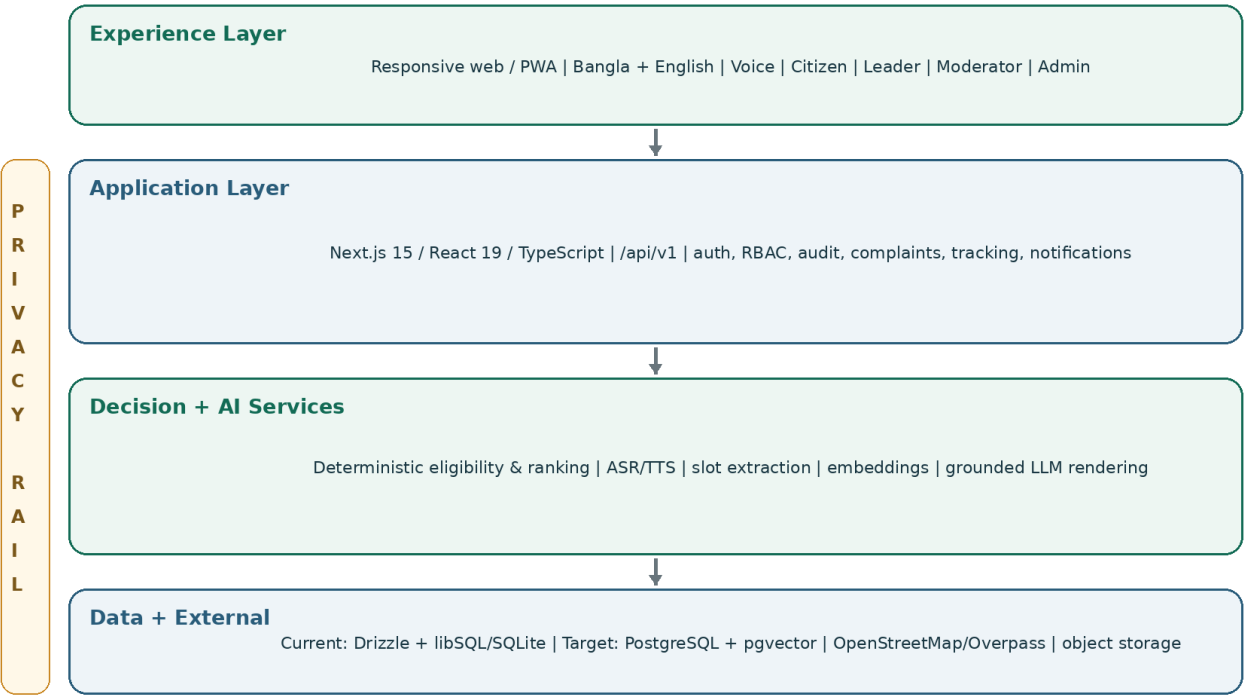


Figure 3. Current modular-monolith foundation with a clear path to scalable, in-country AI services.

6.1 What is built today

The AccessAI feature inventory is based on a working prototype: Next.js 15, React 19 and TypeScript; Drizzle ORM over libSQL/SQLite; 32 tables; 31 API route files exposing 51 handlers; 685 automated tests; phone/OTP/PIN authentication; role-based administration; deterministic eligibility; action plans; notifications; OpenStreetMap nearby services; and AI-provider adapters. The current repository discovery catalogued 219 capabilities across 19 domains. These implementation figures are code-derived and should be treated as the prototype baseline, not projected scope.

6.2 Production path

Component	Deployment approach
Application	2+ stateless Next.js instances behind managed load balancing; REST /api/v1; background job runner for indexing, reminders and corpus review.
Database	Migrate libSQL/SQLite to PostgreSQL; pgvector for embeddings; encrypted backups; row-level ownership checks and audit tables.
Inference	In-country inference host for ASR/TTS/extraction; provider abstraction for LLM rendering; CPU-first encoder inference, optional GPU for speech throughput.
Storage	Private object storage for complaint images / source documents; signed access; no public citizen media URLs.
Mapping	Server-proxied OpenStreetMap tiles / Overpass lookups with caching, attribution and graceful fallbacks.
Observability	Structured request IDs, latency/error metrics, AI model/version logs, grounding failures, data freshness and moderation queues.

7. Risks, Challenges & Responsible AI

The dominant risk is not that the model sounds bad. It is that the system confidently gives a citizen incorrect guidance about a public entitlement. The architecture therefore prioritizes abstention, provenance, human confirmation and reversible operations.

Risk	Impact	Mitigation
Hallucinated or unsupported advice	High	LLM receives a fixed ResponsePlan; no model-generated programme/threshold; citations required; deterministic composer fallback.
Outdated programme rule	High	Source/version/freshness metadata; scheduled staleness review; low confidence for unverified/outdated records; human verification before authoritative use.
ASR / dialect bias	High	Subgroup evaluation by district/age/gender; critical-slot accuracy; transcript read-back; typed/assisted mode remains equivalent.
Wrong extracted citizen fact	High	Per-field provenance, confidence and confirmation; positive eligibility uses confirmed critical fields; abstain rather than guess.
Prompt injection from documents	Medium	Retrieved text treated as data, not instruction; schema-constrained extraction; validation + human review before publication.
Privacy / sensitive data exposure	High	Data minimisation, purpose-based consent, RBAC, encrypted storage, in-country inference option, no retained voice by default.
Institutional non-adoption	High	Integrate with existing channels rather than replace them; phased pilots with UDC/NGO/government partners; clear authority over official rules.
Cost / sustainability	Medium	Deterministic path, open geospatial data, self-host option, batch AI, institutional payer model; measure verification labour before scaling.

7.1 Transparency, accountability, privacy and fairness safeguards

- **Transparency:** show source, verification status, decisive eligibility reason, uncertainty and what the system understood from the citizen.
- **Accountability:** store rule version, profile snapshot, evaluation trace, model/version logs and administrative audit events so a decision can be reconstructed.
- **Privacy:** health and other sensitive data are opt-in; prompts contain only decision-relevant fields; voice is transcribed and discarded by default.
- **Fairness:** missing data returns unknown rather than false; AI cannot override statutory rules; performance is evaluated by subgroup and low-confidence outputs abstain.
- **Human remedy:** citizens can report incorrect information; moderators review content; officials retain final authority over government rules and service decisions.

AI governance stance

The platform advises and guides; it does not autonomously approve benefits, predict a citizen's "worthiness", score fraud risk from language, or submit claims without visible human control.

8. Revenue, Distribution & Sustainability

Citizens should not pay for guidance to public services. The commercial model therefore separates the user from the payer: citizens access the core journey free, while institutions pay for deployment, verified content operations, analytics, integrations and managed infrastructure.

8.1 Institutional revenue model

Payer / line	Value exchanged
Government / local administration	Deployment and integration contracts; administrator dashboards; verified programme/rule management; service-demand and complaint analytics.
NGOs / development partners	Programme-specific deployments, field-worker tools, beneficiary navigation, evidence and reporting, verified knowledge operations.
API / platform consumers	Metered or contracted access to verified programme discovery, eligibility evaluation and nearby-service APIs, subject to governance.
Managed AI hosting	Optional in-country ASR/TTS/embedding and document-intelligence hosting for organizations that cannot use cross-border AI APIs.
Explicitly refused	No sale of citizen data; no referral commission that could bias which programme is recommended; no paywall for basic citizen guidance.

8.2 Distribution strategy

1. Assisted first: Union Digital Centre operators, NGO caseworkers and local service desks use the platform with citizens who are least likely to discover an app independently.
2. Direct second: mobile-responsive web/PWA, voice, Bangla UI and SMS/notification re-engagement for citizens who can self-serve.
3. Integrate, do not duplicate: connect to MyGov/333 and agency service links where available, positioning Shebar Janala as the discovery, eligibility and accountability layer rather than a replacement portal.
4. Government intelligence: aggregate only necessary, privacy-preserving demand/complaint analytics to show where citizens are getting stuck and which programme information is stale.

8.3 Sustainability learning from the pilot

The two-district pilot showed that a high-touch civic reporting model can create engagement but is expensive to operate. The current design changes the economics in three ways: (1) AI and deterministic automation reduce repeated advisory work, (2) existing distribution partners reduce field acquisition cost, and (3) institutional contracts pay for the verified information and operational tooling that citizens cannot reasonably finance. Cost feasibility is now a measurable product requirement, not an assumption.

Commercial north star

Scale with institutions and verified programme coverage, not with citizen fees or attention monetisation.

9. Roadmap, Expected Outcomes & Conclusion

9.1 Roadmap with measurable exit gates

Phase	Scope	Exit gate
Phase 0 - Evidence [done]	Shebar Janala pilot in 2 districts; 33 citizens; 102 complaints. Core AccessAI prototype with deterministic eligibility, retrieval, tracking, maps, admin and safety controls.	Evidence that citizen reporting and service-navigation pain are real; technical baseline exists.
Phase 1 - AI hardening [next]	Domain-adapted Bangla ASR/TTS; slot extraction; embeddings; 10+ verified programmes; document-intelligence review workflow; pilot evaluation set.	Critical-field precision ≥ 0.98 ; ASR slot accuracy reported by district; semantic retrieval improves Recall@10 without harming Precision@3.
Phase 2 - Institutional pilot	Deploy with one NGO / UDC / government partner; integrate real programme data and assisted service workflow; monitor complaint routing and session time.	Measured reduction in unknown eligibility, assisted-session time and information-search effort; programme freshness SLA achieved.
Phase 3 - Scale	Expand programme coverage and districts; API integrations; in-country managed inference; government dashboards and governance agreements.	Multiple institutional consumers, repeatable verification operations and sustainable contract-based deployment.

9.2 Expected results

- Citizens discover relevant government support from a life event rather than a programme name.
- Eligibility guidance becomes consistent, explainable and auditable; missing data produces a question rather than a silent exclusion.
- Voice and read-aloud reduce literacy and typing barriers, while OpenStreetMap connects advice to real service locations.
- Officials receive structured, location-aware complaints and aggregate demand signals rather than disconnected anecdotes.
- Programme-data verification becomes a managed workflow with measurable freshness and reviewer effort.

9.3 Conclusion

Shebar Janala is not a proposal to put an LLM in front of government data. It is a deliberately constrained AI system built around the reality that citizens speak in stories while government services operate through rules. AccessAI translates between those worlds: statistical models understand speech and language; deterministic logic protects eligibility decisions; grounded generation makes the result understandable; and Shebar Janala carries the citizen forward into action and accountability. The first pilot gave us the most valuable evidence a civic-tech team can receive: engagement alone is not enough if the operating model cannot scale. The current architecture is the response to that lesson - more useful to citizens, safer for government, measurable for evaluators and financially more credible for deployment. Our goal for BCOLBD 2026 is to demonstrate not only an AI feature, but a responsible public-service intelligence layer that Bangladesh can realistically adopt.

Submission claim

Working foundation today. Domain-adapted civic AI under development. Measurable pilot evidence. Clear safety boundaries. A credible path from prototype to institutionally deployed GovTech.

Appendix A. Prototype Evidence & Implementation Status

The purpose of this appendix is to separate what exists from what is proposed. Competition judges should be able to reproduce the core logic without relying on narrative claims.

219	19	51	32	685
capabilities catalogued	feature domains	API handlers	database tables	automated tests

A1. Current implementation baseline

Area	Status	Evidence / boundary
Authentication & session	Implemented	Phone/PIN/OTP flows, refresh rotation, logout, session management.
Citizen profile	Implemented	Structured eligibility profile, consent and display preferences.
Conversational assistant	Implemented / mixed	Bilingual conversation, deterministic NLU, grounded response generation, feedback.
Eligibility engine	Implemented	Deterministic rule engine, traces, what-if analysis, replayable audit records.
Discovery & ranking	Implemented / partial	BM25, metadata filters, ranking; embedding channel built but provider-dependent.
Application tracking	Implemented	Saved programmes, 8-stage tracker, action plans, tasks and timeline.
Nearby services	Implemented	OpenStreetMap map, Overpass lookup, distance ordering and provenance badges.
Voice	Implemented / partial	Browser speech where available, server transcription adapter, read-aloud and voice navigation.
Administration	Implemented	Programme/rule management, verification workflow, moderation, analytics and audit.
Document intelligence	Proposed	OCR + layout + constrained rule extraction into human review queue.
Domain-adapted civic models	In development	Fine-tuned speech / extraction / retrieval stack with owned evaluation and release gates.

A2. Repository and live prototype



Code repository

<https://github.com/i-am-mushfiq/AccessAI>



Live leader portal

<https://amar-shebar-janala.vercel.app/leader>

Scope discipline for the AI category

The earlier whitepaper explored a blockchain-based trust registry. It is intentionally removed from the core of this AI-category submission. The AI thesis can be evaluated independently: voice/language understanding, safe information extraction, semantic discovery, deterministic eligibility and grounded explanation. Cryptographic attestation may be revisited later as a separate trust-infrastructure problem, but it is not required for the claimed AI value.

Appendix B. Responsible AI Evaluation & MLOps Protocol

Responsible AI evaluation and release loop

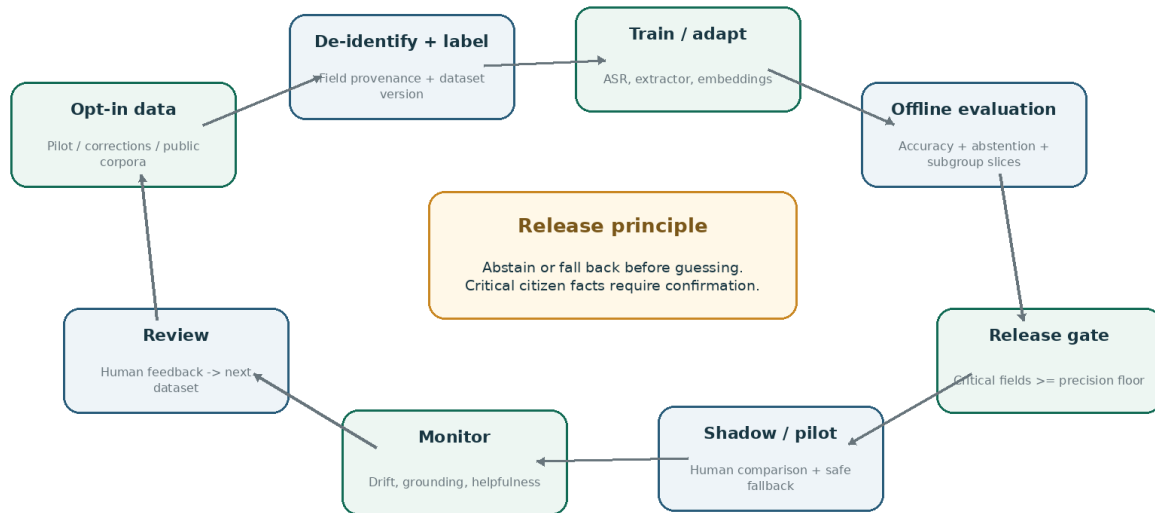


Figure 4. Model development and release loop. The release gate is task-specific and favors abstention over unverified confidence.

B1. Evaluation matrix

Component	Primary metrics	Release interpretation
ASR	WER, CER, critical-slot accuracy	Report overall and by district, age band, gender, device/noise condition. Critical numbers must be tracked separately.
Slot extraction	Precision, recall, F1, abstention	Critical fields target ≥ 0.98 precision. Low confidence $\rightarrow$ ask, do not write.
Retrieval	Recall@10, Precision@3, nDCG@10	Compare BM25-only vs hybrid. Metadata filters remain hard constraints.
Eligibility	Agreement with human-authored rule tests	100% deterministic unit tests for encoded rules; missing data $\rightarrow$ unknown.
Document AI	Field exact match + golden-profile verdict agreement	Proposed rule must pass schema + smoke test + human review before verification.
Grounded generation	Citation coverage, grounding failure, factual consistency	Compare generated reply against ResponsePlan; sample human review; fallback composer on model/provider failure.
Fairness / accessibility	Subgroup error and task-completion gap	Release blockers are subgroup-specific; do not hide poor rural/dialect performance behind an average.

B2. Data governance rules

- Purpose-specific consent: profile data for eligibility is distinct from consent to use corrections/utterances for model improvement.
- Data minimisation: only fields used in the decision are sent to a hosted renderer; citizen-specific generations are not shared caches.
- Voice privacy: audio is transcribed in-request and discarded by default; research retention requires separate consent and retention policy.
- Dataset/model versioning: every deployed model is linked to dataset revision, evaluation report, prompt/schema version and rollback path.
- Human appeal: incorrect information and complaint routing remain reviewable; no opaque model output is treated as an irreversible government decision.

Appendix C. Author, Traction & Product Evolution

C1. Author

Author	Role	Responsibility
Tanjim Mahmud Rakin	Author	Product vision, civic-tech strategy, research, governance, partnerships and whitepaper authorship.

C2. Traction

Milestone	What it proves
Bangladesh ICT & Innovation Award 2025	National Champion, Public Sector category - external validation of the original Shebar Janala civic-tech concept.
APICTA 2025, Taiwan	Represented Bangladesh in the Digital Government context, providing international feedback on scalability and positioning.
Field pilot	Piloted across two districts; 33 citizens engaged; 102 complaints captured. The pilot directly informed the commercial and architectural pivot.
CityGen Builders	Selected for the EU-supported CityGen Builders programme, strengthening the project's civic innovation network.
2026 product pivot	AccessAI integrated as a core Shebar Janala feature; team is developing a domain-adapted Bangla civic AI stack and a more sustainable institutional operating model.

C3. Product evolution

2025: civic reporting + budget transparency -> field pilot -> cost/sustainability learning -> 2026: Shebar Janala becomes an end-to-end service-access and accountability platform, with AccessAI as the intelligence layer -> next: institutionally verified programme data, domain-adapted Bangla models and assisted-service deployments.

What we learned

The project became stronger when we stopped treating “more features” as progress. The key constraints are now explicit: verified programme data, Bangla speech quality, institutional integration, measurable safety, and an operating model someone other than the citizen can sustainably fund.

C4. Demo moments recommended for judges

5. Speak an unscripted Bangla life event; show the transcript and confirmation of a critical field.
6. Show hybrid search retrieving a programme where a paraphrase has little lexical overlap.
7. Open the eligibility trace and prove the LLM did not create the verdict.
8. Generate an action plan and show the nearest relevant service point on OpenStreetMap.
9. Create a service-failure complaint in Shebar Janala and show how the journey continues into accountability.

Appendix D. References

- [1] Blockchain Olympiad Bangladesh. “Guideline & Evaluation Scheme for Artificial Intelligence Category - BCOLBD 2026.” Official competition guideline.
- [2] General Economics Division / Social Security Policy Support Programme, Government of Bangladesh. “Strategic Framework of the National Social Security Strategy 2026-2035.” 2026.
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- [14] AccessAI repository and feature inventory, branch feature/openstreetmap-nearby. Implementation-derived inventory used for current-state claims.

Submission links

Live prototype: <https://amar-shebar-janala.vercel.app/leader>

Repository: <https://github.com/i-am-mushfiq/AccessAI>

Competition rules / guideline: <https://bcolbd.org/rules>

Document note

Prepared for Blockchain Olympiad Bangladesh 2026, Artificial Intelligence Category. Main whitepaper is 10 pages; appendices provide implementation evidence, evaluation protocol, author/traction and references. All pilot metrics are reported as observed project data; future performance targets are explicitly framed as release gates rather than achieved outcomes.